

20-bit Delta-Sigma ADC: Modulator & Decimation Filter Design

Mid Progress Report | EMBED Club | 4 July 2026

1. Conceptual Understanding

- **Regular ADCs & sampling:** Nyquist-rate ADC architectures such as Flash and SAR exhibit architecture-dependent trade-offs among resolution, sampling rate, power consumption, area, and complexity.
- **SQNR, ENOB & oversampling:** For an ideal ADC, $SQNR \approx 6.02N + 1.76$ dB. Practical ENOB is commonly estimated from measured SINAD using $ENOB = (SINAD - 1.76)/6.02$. The oversampling ratio is defined as $OSR = f_s/(2B)$, where f_s is the modulator sampling frequency and B is the signal bandwidth.
- **Delta-Sigma modulator:** pushes quantization noise out of the signal band using a feedback integrator and quantizer loop. For an ideal L -th-order modulator, the noise transfer function behaves approximately as $NTF(z) \approx (1 - z^{-1})^L$, providing progressively stronger suppression of in-band quantization noise with increasing modulator order. The complete $\Delta\Sigma$ ADC chain consists of oversampling, noise shaping, digital filtering, and decimation.
- **Role of the digital/decimation filter:** Digital filters such as CIC/SINC, FIR, and IIR filters attenuate out-of-band shaped quantization noise before decimation, preventing aliasing as the sampling rate is reduced.

2. Experimentation So Far

Built behavioral models of 1st-order and 2nd-order single-bit Delta-Sigma modulators and swept $OSR \in \{2^4 \dots 2^8\}$ using an identical coherent half-scale sinusoidal test tone, with the half-scale input chosen to provide operating margin against modulator overload. After discarding startup transients, estimated in-band ENOB was obtained from a coherent 2^{18} -sample FFT by calculating in-band SINAD and applying $ENOB = (SINAD - 1.76)/6.02$.

- Estimated ENOB increases approximately linearly with $\log_2(OSR)$. The measured 1st-order and 2nd-order models achieve approximately 1.35 and 2.48 bits of ENOB improvement per OSR doubling, respectively, compared with ideal theoretical trends of approximately 1.5 and 2.5 bits/octave.
- With its steeper noise-shaping slope, the 2nd-order single-bit modulator yields approximately $6.4 \rightarrow 16.3$ estimated ENOB over $OSR 16 \rightarrow 256$, crossing the 16-bit target near $OSR = 256$, while the 1st-order model remains below the target within this sweep.
- The 2nd-order modulator is therefore the current leading candidate for further evaluation toward the 16–19 bit ENOB target. The design target is a nominal 20-bit digital output, while the practical performance objective is an ENOB of 16–19 bits.

- **Current limitations:** Results are limited to behavioral single-tone simulations without explicit decimation filtering or circuit-level non-idealities; multi-tone, broadband, and filter-inclusive validation form the next stage.

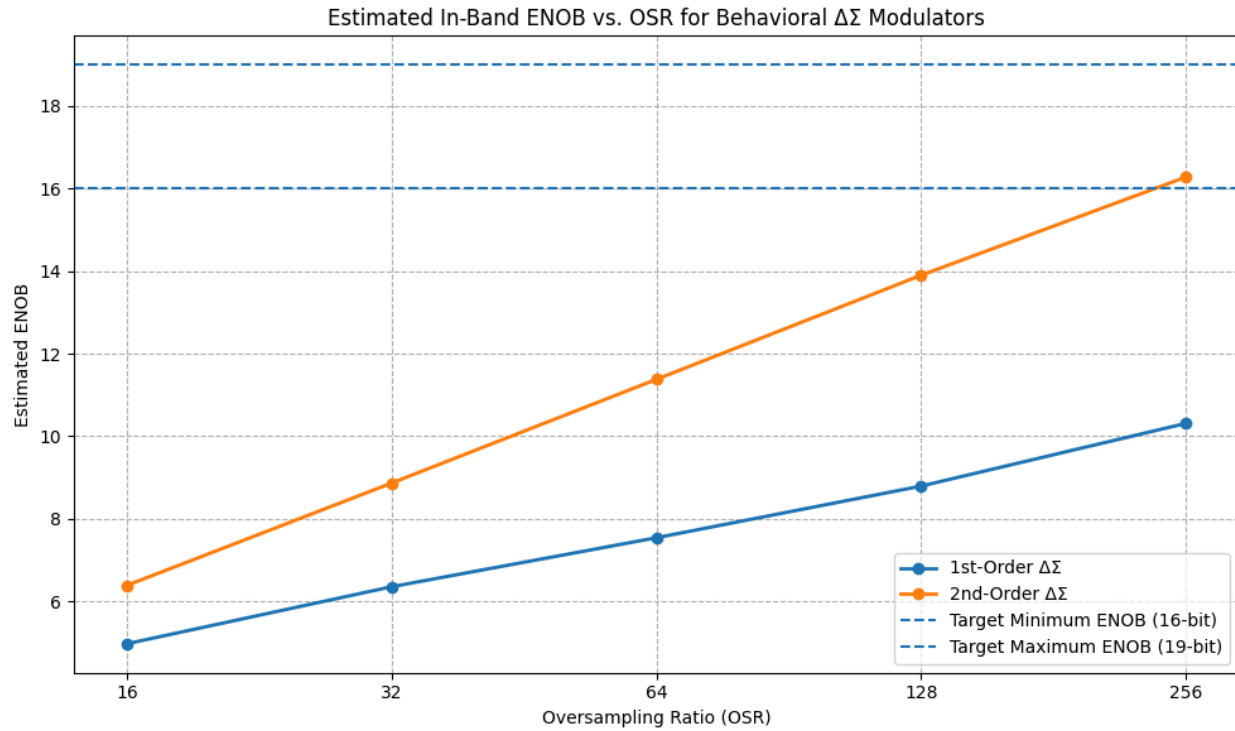


Fig. 1 — Estimated in-band ENOB vs. OSR for behavioral 1st- and 2nd-order single-bit $\Delta\Sigma$ modulators, compared against the 16–19 bit target range.

3. Where I'm Headed

- **Step 5 (in progress):** Validate the current 2nd-order modulator by re-running the OSR sweep with representative low-frequency single-tone, multi-tone, and broadband test signals corresponding to the target 0.5–2 kps Nyquist output sampling rates. The relationship between output sampling rate, OSR, and required modulator clock frequency will also be evaluated; for example, a 1 kps output rate at OSR = 256 requires a modulator sampling frequency of approximately 256 kHz.
- **Step 6 (ideation):** Evaluate SINC/CIC and FIR-based decimation architectures. The current leading candidate is a multistage CIC/SINC³ coarse decimator followed by a short FIR compensation filter because of its multiplier-free first stage and RTL suitability. Architecture selection will be based on ENOB preservation, required OSR and clock rate, passband droop, stopband attenuation, and RTL implementation complexity.
- **Next :** Evaluate whether a third-order single-loop or cascaded MASH architecture provides sufficient ENOB improvement to justify its additional stability and implementation complexity. The selected decimation filter will then be translated to synchronous Verilog RTL and verified bit-for-bit against reference test vectors, followed by FFT-based comparison of the RTL output with the behavioral-model baseline.